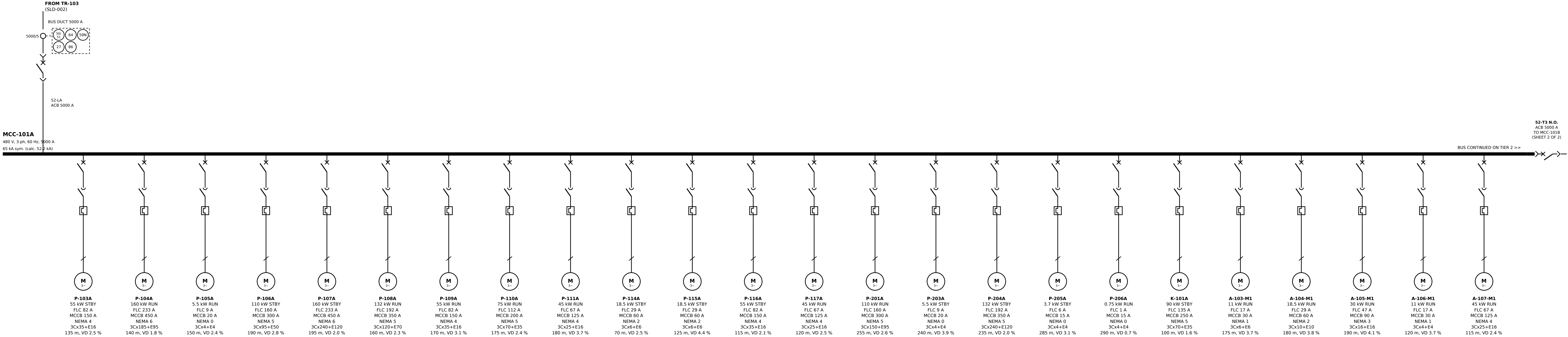
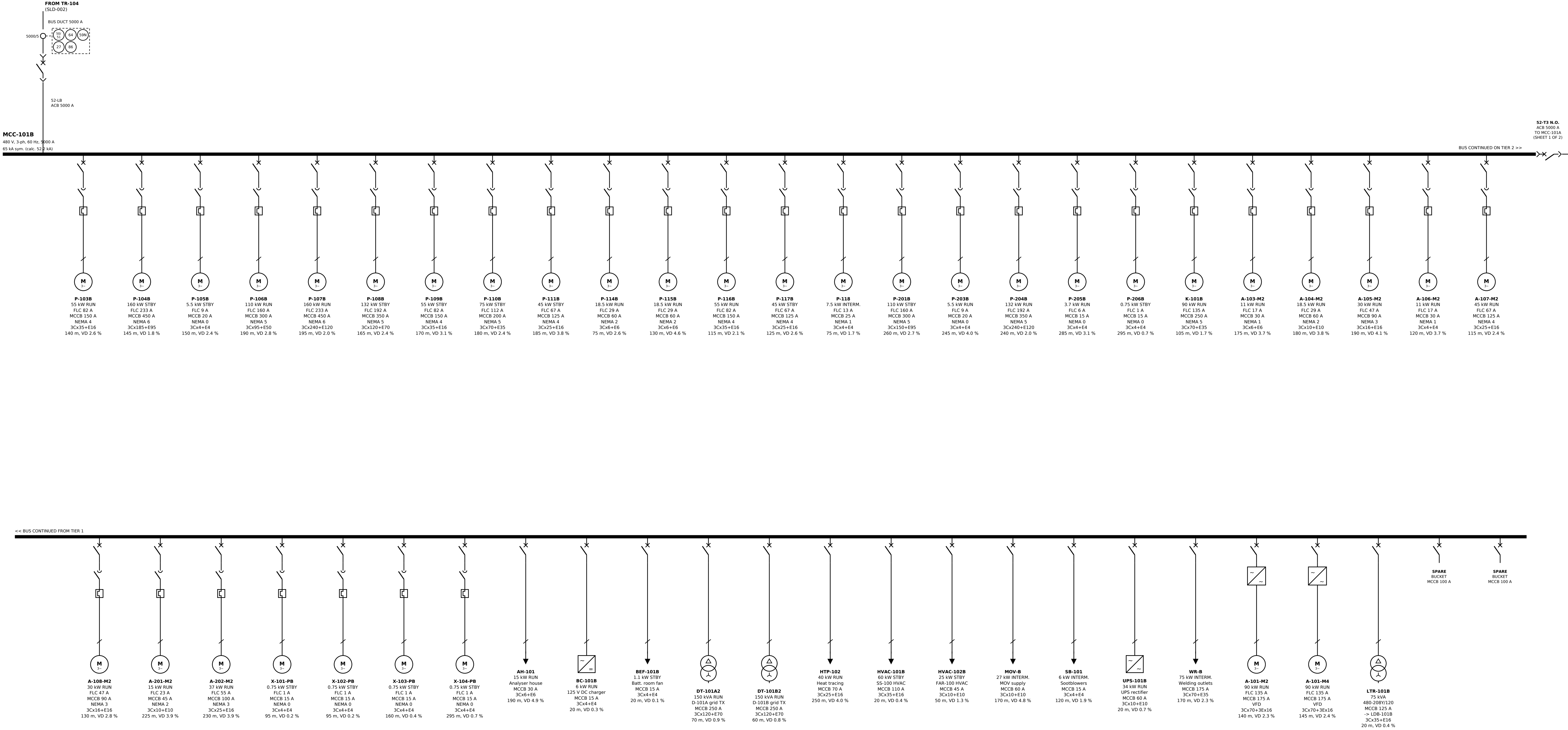


MCC-101A SUMMARY	VALUE
Feeders / buckets	48 (33 motor, 2 spare)
Connected load	1,768 kW
Continuous / Interm. / Standby	1,082 / 127 / 559 kW
Maximum demand (C+0.3I+0.1S)	1,176 kW / 1,340 kVA, PF 0.88
Both sections on one TR (tie closed)	2,673 kVA vs TR 3000/3750 kVA ONAN/ONAF
Largest DOL motor start - bus dip	3.5 % (limit 10 %)

- NOTES**
- MCC-101A: 480 V, 3-ph 3-W, 5000 A Cu bus, 65 kA sym.; ACB incomer/tie in LV switchgear section (IEEE C37.20.1), MCC sections UL 845, withdrawable buckets.
 - Motor buckets: MCCB (thermal-magnetic, <= 250 % FLC per NEC 430.52) + contactor + electronic OL, IEC 60947-4-1 Type 2 coordination; 120 V control transformer; DCS hardwired start/stop/run/trip.
 - Motors >= 30 kW; 4-20 mA motor current to DCS. VFDs: 6-pulse + passive filter (THDI <= 8 %).
 - Cable: Cu/XLPE/SWA/PVC 0.6/1 kV (IEC mm2). Sizes per CFU-000-EL-CBL-001 (ampacity x 1.25 FLC, VD <= 5 % running, <= 15 % at motor terminals during start, 5C withstand via MCCB let-through).
 - Pump A/B pairs on opposite MCC sections; RUN/STBY = normally running unit for demand calculation.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
SINGLE LINE DIAGRAM 480 V SWITCHGEAR / MCC-101A		
DISCIPLINE	STAGE	SCALE
ELECTRICAL	FEED	NTS
SHEET		1 OF 2
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DATE		2026-10-02
DRAWING NUMBER		REV
CFU-000-EL-SLD-003		A



MCC-101B SUMMARY	VALUE
Feeders / buckets	49 (34 motor, 2 spare)
Connected load	1.778 kW
Continuous / Interm. / Standby	1.079 / 124 / 575 kW
Maximum demand (C+0.3I+0.1S)	1.173 kW / 1.333 kVA, PF 0.88
Both sections on one TR (tie closed)	2.673 kVA vs TR 3000/3750 kVA ONAN/ONAF
Largest DOL motor start - bus dip	3.5 % (limit 10 %)

NOTES

1. MCC-101B: 480 V, 3-ph 3-W, 5000 A Cu bus, 65 kA sym.; ACB incomer/tie in LV switchgear section (IEEE C37.20.1), MCC sections UL 845, withdrawable buckets.

2. Motor buckets: MCCB (thermal-magnetic, <= 250 % FLC per NEC 430.52) + contactor + electronic OL, IEC 60947-4-1 Type 2 coordination; 120 V control transformer; DCS hardwired start/stop/run/trip.

3. Motors >= 30 kW; 4-20 mA motor current to DCS. VFDs: 6-pulse + passive filter (THDI <= 8 %).

4. Cable: Cu/XLPE/SWA/PVC 0.6/1 kV (IEC mm2). Sizes per CFU-000-EL-CBL-001 (ampacity x 1.25 FLC, VD <= 5 % running, <= 15 % at motor terminals during start, 5C withstand via MCCB let-through).

5. Pump A/B pairs on opposite MCC sections; RUN/STBY = normally running unit for demand calculation.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
SINGLE LINE DIAGRAM 480 V SWITCHGEAR / MCC-101B		
DISCIPLINE	STAGE	SCALE
ELECTRICAL	FEED	NTS
SHEET		2 OF 2
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DATE		2026-10-02
DRAWING NUMBER		REV
CFU-000-EL-SLD-003		A